EE2023 Midterm Cheat Sheet

By: Brians Tjipto

1. Basic Continuous-Time Signals

Rectangular Pulse: $\mathrm{rect}(t) = 1$ for $|t| < 0.5$, 0 otherwise.
Triangular Pulse: $\mathrm{tri}(t) = 1 - |t|$ for $|t| < 1$, 0 otherwise.
Unit Step: $u(t) = 1$ for $t > 0$, 0 for $t < 0$.
Signum: $\mathrm{sgn}(t) = 1$ for $t > 0$, -1 for $t < 0$.
Sinc Function (Normalized): $\mathrm{sinc}(t) = \frac{\sin(\pi t)}{\pi t}$. *Note:* $\mathrm{sinc}(0) = 1$, *and crosses zero at integer values* $t \neq 0$.
Exponential: $e^{-\alpha t}u(t)$ (Right-sided decaying, $\alpha > 0$).
From a Mag and Phase Spectrum: Magnitude $\cdot\, e^{j(2\pi f t + \mathrm{Phase})}$
Half-Wave Rectification: To chop off the negative bottom of a signal $x(t)$, multiply it by: $\frac{1}{2}\{\mathrm{sgn}[x(t)] + 1\}$ or use $\frac{1}{2}\{x(t) + |x(t)|\}$.
Building $x(t)$ from discrete spectra: For each spike at frequency f, the time-domain term is Magnitude $\cdot\, e^{j(2\pi f t + \mathrm{Phase})}$.

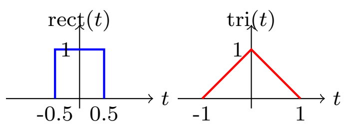
2. Time-Domain Operations (Graphical)

Key Rule (Inverse Operations): Transformations inside the function argument act *inversely* on the visual graph's time axis:
- **Shift Right:** Subtract inside the equation (e.g., $t - b$).
- **Shift Left:** Add inside the equation (e.g., $t + b$).
- **Compress (Divide R):** Multiply inside the equation (e.g., at).
- **Expand (Multiply R):** Divide inside the equation (e.g., t/a).

For a given signal $x(t)$, a combined transformation $y(t) = A \cdot x(at - b)$ follows a strict order of operations if done visually:
1. Time Shift by b (right if $b > 0$, left if $b < 0$).
2. Time Scale by a (compress if $|a| > 1$, expand if $|a| < 1$).
3. Time Reverse if a is negative (flip over $t = 0$).
4. Amplitude Scale by A (stretch vertically).

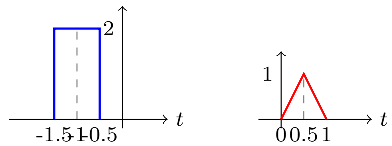
Zero-Mapping: To find where a specific point of the original signal $x(t)$ ends up in the new signal $y(t) = x(at - b)$, set the new argument equal to the original time point. *Example:* If $\mathrm{rect}(t)$ is centered at $t = 0$, the center of $\mathrm{rect}(at - b)$ is at $at - b = 0 \implies t = b/a$.

Base Signals: $\mathrm{rect}(t)$ and $\mathrm{tri}(t)$

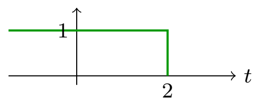


Amplitude & Time Shift: $2\,\mathrm{rect}(t+1)$ **BLUE**
$b = -1 \implies$ Shift left by 1. $A = 2 \implies$ Amplitude scaled to 2.
Center: $t + 1 = 0 \implies t = -1$.
Edges: $t + 1 = \pm 0.5 \implies t = -1.5$ and $t = -0.5$.

Time Scale & Shift: $\mathrm{tri}(2t - 1)$ **RED**
$b = 1 \implies$ Shift right by 1. $a = 2 \implies$ Compress by 2.
Center: $2t - 1 = 0 \implies t = 0.5$.
Edges: $2t - 1 = \pm 1 \implies t = 0$ and $t = 1$.
Width: Original width is $2 \implies$ New width is $2/2 = 1$.



Time Reversal & Shift: $u(2 - t) \equiv u(-t + 2)$
This is the unit step $u(t)$ flipped and shifted.
Edge transition: $2 - t = 0 \implies t = 2$.
Since $u(t) = 1$ for positive arguments, $u(2 - t) = 1$ when $2 - t > 0 \implies t < 2$.



3. The Dirac Impulse $\delta(t)$

Definition: $\delta(t) = 0$ for $t \neq 0$, and $\int_{-\infty}^{\infty} \delta(t)dt = 1$.
Sifting Property: $\int_{-\infty}^{\infty} x(t)\delta(t - t_0)dt = x(t_0)$.
Multiplication: $x(t)\delta(t - t_0) = x(t_0)\delta(t - t_0)$.
Time Scaling: $\delta(at - b) = \frac{1}{|a|}\delta(t - \frac{b}{a})$.
Comb Function: $\mathrm{comb}_T(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT)$.
Spike Weight $= f_0 \cdot G(kf_0)$.

4. Energy and Power Signals

Total Energy E: $E = \int_{-\infty}^{\infty} |x(t)|^2 dt$.
Average Power P: $P = \lim_{T\to\infty} \frac{1}{2T}\int_{-T}^{T} |x(t)|^2 dt$.
For periodic signals: $P = \frac{E_{\text{one period}}}{T}$
Energy Signals: $0 < E < \infty$ and $P = 0$ (e.g., transients).
Power Signals: $0 < P < \infty$ and $E = \infty$ (e.g., periodic, constants).
An Energy Signal has a finite area under its squared curve, meaning it eventually starts and stops. A Power Signal repeats forever with infinite energy. If $x(t)$ is entirely flat at 0 outside the window $[-lower, upper]$, its total energy integral is finite ($< \infty$). Thus, it is an Energy Signal.
Energy of a Triangle: $\int_{-\infty}^{\infty} |\mathrm{tri}(t/T)|^2 dt = \frac{2T}{3}$. (Use for area under squared magnitude spectra).
Energy of a Sinc: $\int_{-\infty}^{\infty} |A\,\mathrm{sinc}(Wt)|^2 dt = \frac{A^2}{W}$. (Derived via Parseval's area of a squared rect).
Transformations on Energy & Power
Given a signal $x(t)$ with energy E_x and power P_x, a transformed signal $y(t) = A \cdot x(at - b)$ will have new energy and power according to these shortcut rules:
1. Amplitude Scaling (A): Both E and P scale by the square of the amplitude because they integrate the squared signal. (Multiplier: $\times|A|^2$)
2. Time Scaling (a):

- **Energy:** Compressing the signal ($|a| > 1$) reduces the area it takes up, expanding ($|a| < 1$) increases it. (Multiplier: $\div|a|$)
- **Power:** Stretching a periodic signal makes the pulses wider, but it also stretches the gap between them proportionally. The ratio of "signal" to "empty space" remains identical. (Multiplier: $\times 1$, no effect on average power)

3. Time Shift (b) & Reversal ($-$): Sliding the signal left/right or flipping it horizontally does not change its total area or average power. (Multiplier: $\times 1$, no effect)
General Formulas:

$$E_y = \frac{|A|^2}{|a|}E_x, \quad P_y = |A|^2 P_x$$

Examples:

- **Energy:** $w(t) = 3x(-10t + 8)$.
  Let $E_x = 40$ J.
  $A = 3, a = -10 \implies E_w = \frac{3^2}{|-10|} \times 40 = \frac{9}{10} \times 40 = 36$ Joules.
- **Power:** $z(t) = \frac{1}{2}y(-\frac{t}{3} + 5)$.
  Let $P_y = 10$ W.
  $A = \frac{1}{2}, a = -\frac{1}{3} \implies P_z = \left(\frac{1}{2}\right)^2 \times 10 = \frac{1}{4} \times 10 = 2.5$ Watts.

5. Fourier Series (FS) - Discrete Spectrum

Fourier Series is strictly reserved for periodic signals that repeat over and over. If $x(t)$ is a one-time, aperiodic event, it cannot be represented by a Fourier Series . For a periodic signal $x_p(t)$ with period T_p and fundamental frequency $f_p = 1/T_p$
Complex Exponential FS:

$$x_p(t) = \sum_{k=-\infty}^{\infty} c_k e^{j2\pi k f_p t}$$

Coefficients c_k:

$$c_k = \frac{1}{T_p}\int_{-T_p/2}^{T_p/2} x_p(t)e^{-j2\pi k f_p t}dt$$

Properties of c_k: If $x(t)$ is real, $c_{-k} = c_k^*$.
If $x(t)$ is real and EVEN, c_k is real and even.
If $x(t)$ is real and ODD, c_k is purely imaginary and odd.
DC Component $= c_0 = \frac{1}{T_p}\int_{T_p} x(t)dt$ (Average value). For $-T_p/2$ to $T_p/2$ (or 0 to T_p)
Parseval's Theorem for FS: Total Power $P = \sum_{k=-\infty}^{\infty} |c_k|^2$.
Total average power is simply the sum of the squared magnitudes of all your coefficients. The PSD is just a graph of those squared magnitudes, placed at their actual frequencies ($f = k \times f_p$)

6. Fourier Transform (FT) - Cont.

We are given $x(t) = 10 + 5\cos(2\pi t) - 5\sin(3\pi t) + 5e^{j(5\pi t + \pi/3)}$.
The frequencies attached to t are $2\pi, 3\pi,$ and 5π rad/s.
The fundamental frequency (ω_p) is the Greatest Common Divisor of these, which is π rad/s. In Hertz: $f_p = \omega_p/2\pi = 0.5$ Hz.
Forward Transform (Time to Frequency f in Hz):

$$X(f) = \int_{-\infty}^{\infty} x(t)e^{-j2\pi f t}dt \quad = \mathcal{F}\{x(t)\}$$

Inverse Transform (Frequency to Time):

$$x(t) = \int_{-\infty}^{\infty} X(f)e^{j2\pi f t}df \quad = \mathcal{F}^{-1}\{X(f)\}$$

Dirichlet Conditions: For FT to exist, $x(t)$ must be absolutely integrable ($\int |x(t)|dt < \infty$), have finite maxima/minima, and finite discontinuities.
The harmonic number k is just the frequency divided by the fundamental frequency ($k = f/f_p$).We pair the magnitudes and phases from the graphs to build $c_k = \mathrm{Magnitude} \cdot e^{j\mathrm{Phase}}$.

7. FT Properties

Linearity: $ax_1(t) + bx_2(t) \rightleftharpoons aX_1(f) + bX_2(f)$
Time Shifting: $x(t - t_0) \rightleftharpoons X(f)e^{-j2\pi f t_0}$ (Adds linear phase)
Frequency Shifting: $x(t)e^{j2\pi f_c t} \rightleftharpoons X(f - f_c)$ (Modulation)
Time Scaling: $x(at) \rightleftharpoons \frac{1}{|a|}X\left(\frac{f}{a}\right)$ (Time compress = Freq expand)
Duality: $X(t) \rightleftharpoons x(-f)$
Time Differentiation: $\frac{d^n}{dt^n}x(t) \rightleftharpoons (j2\pi f)^n X(f)$
Time Integration: $\int_{-\infty}^{t} x(\tau)d\tau \rightleftharpoons \frac{X(f)}{j2\pi f} + \frac{1}{2}X(0)\delta(f)$
Convolution in Time: $x_1(t) * x_2(t) \rightleftharpoons X_1(f)X_2(f)$
Multiplication in Time: $x_1(t)x_2(t) \rightleftharpoons X_1(f) * X_2(f)$
Area / Integral Trick: $\int_{-\infty}^{\infty} x(t)dt = X(0)$. (Evaluates complex integrals instantly)
Parseval's (Rayleigh's) Theorem:
$\int_{-\infty}^{\infty} |x(t)|^2 dt = \int_{-\infty}^{\infty} |X(f)|^2 df$ Replicating a signal every 4 seconds ($T = 4$) takes its continuous frequency spectrum $X(f)$ and slices it into discrete spikes spaced by $f_0 = 1/4$ Hz.The formula applies a $1/T$ scale factor: $Y(f) = \sum \frac{1}{4}X\left(\frac{k}{4}\right)\delta\left(f - \frac{k}{4}\right)$.
Plug in our $X(f) = \text{sinc}(2f) + \text{sinc}^2(f)$:
$Y(f) = \sum_{\mathbf{k}=-\infty}^{\infty} \frac{\mathbf{1}}{\mathbf{4}}\left[\text{sinc}\left(\frac{\mathbf{k}}{\mathbf{2}}\right) + \text{sinc}^{\mathbf{2}}\left(\frac{\mathbf{k}}{\mathbf{4}}\right)\right]\delta\left(\mathbf{f} - \frac{\mathbf{k}}{\mathbf{4}}\right)$.

$\text{tri}\left(\frac{t}{T}\right) = \begin{cases} 1 - \frac{|t|}{T}, & |t| \le T \\ 0, & |t| > T \end{cases}, \text{rect}\left(\frac{t}{T}\right) = \begin{cases} 1, & -T/2 \le t < T/2 \\ 0, & \text{elsewhere} \end{cases}$

8. FT Table (Standard Pairs)

$$\text{rect}(t/T) \rightleftharpoons T\,\text{sinc}(fT)$$
$$\text{tri}(t/T) \rightleftharpoons T\,\text{sinc}^2(fT)$$
$$\delta(t) \rightleftharpoons 1$$
$$1 \rightleftharpoons \delta(f)$$
$$\delta(t - t_0) \rightleftharpoons e^{-j2\pi f t_0}$$
$$e^{j2\pi f_0 t} \rightleftharpoons \delta(f - f_0)$$
$$\cos(2\pi f_c t) \rightleftharpoons \frac{1}{2}[\delta(f - f_c) + \delta(f + f_c)]$$
$$\sin(2\pi f_c t) \rightleftharpoons \frac{1}{j2}[\delta(f - f_c) - \delta(f + f_c)]$$
$$e^{-\alpha t}u(t) \rightleftharpoons \frac{1}{\alpha + j2\pi f} \quad (\alpha > 0)$$
$$te^{-\alpha t}u(t) \rightleftharpoons \frac{1}{(\alpha + j2\pi f)^2}$$
$$\text{sgn}(t) \rightleftharpoons \frac{1}{j\pi f}$$
$$u(t) \rightleftharpoons \frac{1}{2}\delta(f) + \frac{1}{j2\pi f}$$
$$\text{comb}_T(t) \rightleftharpoons f_s\,\text{comb}_{f_s}(f) \quad \text{where } f_s = 1/T$$

Time Scaling: $\mathcal{F}\{x(at)\} = \frac{1}{|a|}X\left(\frac{f}{a}\right)$
Time Shifting: $\mathcal{F}\{x(t - t_0)\} = X(f)e^{-j2\pi f t_0}$
$y(t) = x(at - b) \rightleftharpoons \mathcal{F}\{x(at - b)\} = \frac{1}{|a|}X\left(\frac{f}{a}\right)e^{-j2\pi f\left(\frac{b}{a}\right)}$
$y(t) = x\left(\frac{t - t_0}{c}\right) \rightleftharpoons \mathcal{F}\left\{x\left(\frac{t - t_0}{c}\right)\right\} = |c|X(cf)e^{-j2\pi f t_0}$

9. Spectral Density & Bandwidth

DC value is simply the frequency at exactly 0 Hz. $X(0) = 2e^0 = \mathbf{2}$
Energy Spectral Density (ESD): $E_x(f) = |X(f)|^2$ (Joules/Hz).
Total Energy $E = \int_{-\infty}^{\infty} E_x(f)df$ (Rayleigh Energy Theorem).
Power Spectral Density (PSD): For periodic signals,
$P_x(f) = \sum_{k=-\infty}^{\infty} |c_k|^2\delta(f - kf_p)$ (Watts/Hz).
Total Power $P = \int_{-\infty}^{\infty} P_x(f)df = \sum |c_k|^2$.
Autocorrelation Function $R_x(\tau)$**:** For energy signals:
$R_x(\tau) = \int x(t)x(t + \tau)dt \rightleftharpoons E_x(f)$.
For power signals: $R_x(\tau) = \lim_{T\to\infty} \frac{1}{2T}\int_{-T}^{T} x(t)x(t + \tau)dt \rightleftharpoons P_x(f)$.
Note: $R_x(0) = E$ (for energy signals) or P (for power signals).
Bandwidth (BW) Definitions:
Combined Nulls: For $x * y \implies$ 1st null is the **minimum** of individual nulls. For $x + y$ (if spectra are strictly positive) $\implies$ 1st null is the **LCM** of individual nulls.

- **3dB bandwidth:** The magnitude squared drops to half its peak value.
- **Half-Power (3-dB) BW:** The frequency where the ESD drops to exactly half of its peak value, or where $|X(f)| \ge \frac{1}{\sqrt{2}}|X(f)_{max}|$.
  Example: Find 3dB BW for $E_x(f) = 4e^{-0.2|f|}$
  - Peak is at $f = 0 \implies E_x(0) = 4$. Half of peak is 2.
  - Set equation to half-peak: $4e^{-0.2f} = 2 \implies e^{-0.2f} = 0.5$
  - Take natural log: $-0.2f = \ln(0.5) \implies -0.2f = -\ln(2)$
  - Solve for f: $f = \frac{\ln(2)}{0.2} = \mathbf{5}\ln(\mathbf{2})$ Hz
- **Absolute BW:** The exact range of positive frequencies where $|X(f)| > 0$. Often infinite for time-limited signals.
- **First-Null BW:** The frequency f at which the main lobe of $|X(f)|$ first hits zero. For $\text{rect}(t/\tau)$, First-Null BW $= 1/\tau$ Hz.
- **M% Energy Containment BW:** The smallest band B containing M% of total energy: $\int_{-B}^{B} E_x(f)df = \frac{M}{100}E$ (Lowpass).
- **Modulated (Bandpass) BW:** When a lowpass baseband signal is modulated to a high frequency carrier, its Passband Bandwidth becomes exactly **double** its Baseband Bandwidth.

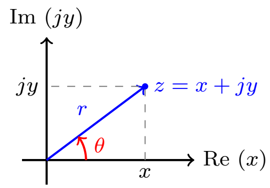
Modulated ESD: If $y(t) = 2x(t)\cos(2\pi f_c t)$ and the shifted spectra do not overlap, the new ESD is simply
$E_y(f) = E_x(f - f_c) + E_x(f + f_c)$.

10. Useful Math & Trig Identities

Euler's: $e^{\pm j\theta} = \cos\theta \pm j\sin\theta$
$j = e^{j0.5\pi}, \quad -j = e^{-j0.5\pi}$
$e^{j\theta} + e^{-j\theta} = 2\cos\theta, \quad \text{sinc}(x) = \frac{\sin(\pi x)}{\pi x}$
$\cos(\theta) = \frac{1}{2}(e^{j\theta} + e^{-j\theta}), \quad \sin(\theta) = \frac{1}{j2}(e^{j\theta} - e^{-j\theta})$
$\cos^2(\theta) = \frac{1}{2}[1 + \cos(2\theta)], \quad \sin^2(\theta) = \frac{1}{2}[1 - \cos(2\theta)]$
$\cos^3(\theta) = \frac{3}{4}\cos(\theta) + \frac{1}{4}\cos(3\theta), \quad \sin^3(\theta) = \frac{3}{4}\sin(\theta) - \frac{1}{4}\sin(3\theta)$
$\sin(\alpha \pm \beta) = \sin\alpha\cos\beta \pm \cos\alpha\sin\beta$
$\cos(\alpha \pm \beta) = \cos\alpha\cos\beta \mp \sin\alpha\sin\beta$
$2\cos\alpha\cos\beta = \cos(\alpha - \beta) + \cos(\alpha + \beta)$
$2\sin\alpha\sin\beta = \cos(\alpha - \beta) - \cos(\alpha + \beta)$
$2\sin\alpha\cos\beta = \sin(\alpha + \beta) + \sin(\alpha - \beta)$

11. Complex Number Forms & Conversions

A complex number z can be represented in three main ways in electrical engineering (using $j = \sqrt{-1}$).
1. Rectangular Form: $z = x + jy$
2. Polar Form: $z = r\angle\theta$
3. Euler (Exponential) Form: $z = re^{j\theta}$



Euler ↔ Polar (Direct Translation):
These are just notation differences. The magnitude r and angle θ remain exactly the same.
$$re^{j\theta} \iff r\angle\theta$$
Polar / Euler → Rectangular:
Given r and θ, find x and y using trigonometry:
$$x = r\cos(\theta), \quad y = r\sin(\theta)$$
$$z = r(\cos\theta + j\sin\theta)$$
Rectangular → Polar / Euler:
Given x and y, find magnitude r and angle θ:
$$r = \sqrt{x^2 + y^2} = |z|$$
$$\theta = \tan^{-1}\left(\frac{y}{x}\right) = \arg(z)$$
Important Quadrant Check for θ*:*

- **Quad I** $(x > 0, y > 0)$: $\theta = \tan^{-1}(y/x)$
- **Quad II** $(x < 0, y > 0)$: $\theta = \tan^{-1}(y/x) + 180^\circ$ (or $+\pi$)
- **Quad III** $(x < 0, y < 0)$: $\theta = \tan^{-1}(y/x) - 180^\circ$ (or $-\pi$)
- **Quad IV** $(x > 0, y < 0)$: $\theta = \tan^{-1}(y/x)$

Operations Guide:

- **Addition/Subtraction:** ALWAYS use Rectangular form. $(x_1 + jy_1) \pm (x_2 + jy_2) = (x_1 \pm x_2) + j(y_1 \pm y_2)$
- **Multiplication:** Use Euler/Polar form. $(r_1e^{j\theta_1}) \times (r_2e^{j\theta_2}) = (r_1r_2)e^{j(\theta_1+\theta_2)}$
- **Division:** Use Euler/Polar form. $\frac{r_1e^{j\theta_1}}{r_2e^{j\theta_2}} = \left(\frac{r_1}{r_2}\right)e^{j(\theta_1-\theta_2)}$

$A\text{sinc}^2(Wt) \rightleftharpoons \frac{A}{W}\text{tri}\left(\frac{f}{W}\right)$ $A\text{sinc}(Bt) \rightleftharpoons \frac{A}{B}\text{rect}\left(\frac{f}{B}\right)$
Every complex exponential $e^{j2\pi f_0 t}$ simply transforms into a Dirac delta spike $\delta(f - f_0)$
Convolution of Rects: $A\,\text{rect}(\frac{t}{T}) * B\,\text{rect}(\frac{t}{T}) = A \cdot B \cdot T \cdot \text{tri}(\frac{t}{T})$.
(Same rule applies for frequency domain f and width W).

Found: https://github.com/brianstm/NUS.git